

```

return templates.TemplateResponse("result.html", {
    "request": request,
    "username": username,
    "user_id": user_id,
    "age": age,
    "weight": weight,
    "goal": goal,
    "intensity": intensity,
    "workout_plan": plan,
    "nutrition_tip": nutrition_tip
})

```

- This allows dynamic plan and nutrition tip display after form submission.

Bind Backend Data to HTML Templates

➤ index.html

- Form elements use `name="..."` to bind to backend `Form(...)` parameters.
- Submits data to `/generate-workout`.

➤ result.html

- Uses:
 - `{{ username }}`, `{{ goal }}`, `{{ intensity }}`, etc. to show user data.
 - `{{ workout_plan }}` and `{{ nutrition_tip }}` to show AI-generated results.
- Includes a feedback form submitted via POST to `/submit-feedback`.

➤ all_users.html

- Loops over all user records using:

```

</tr>
{% for user in users %}
<tr>
    <td>{{ user.id }}</td>
    <td>{{ user.name }}</td>
    <td>{{ user.age }}</td>
    <td>{{ user.weight }}</td>
    <td>{{ user.goal }}</td>
    <td>{{ user.intensity }}</td>
    <td><pre>{{ user.original_plan }}</pre></td>
    <td><pre>{{ user.updated_plan }}</pre></td>
</tr>
{% endfor %}

```

Milestone 5. Deployment

Now, the focus is on deploying the **FitBuddy** application on a local system using **FastAPI** and **Uvicorn**. The app uses **HTML templates rendered with Jinja2**, alongside AI services powered by **Gemini**. This milestone ensures all environment dependencies are correctly installed, and the app runs smoothly on a local server, preparing it for future deployment on cloud platforms.

Activity 5.1: Preparing the Application for Local Deployment

Set up a virtual environment

- Create a virtual environment to manage dependencies and isolate the FitBuddy project from other environments.
- Activate the environment and install all necessary dependencies listed in the [requirements.txt](#) file, including FastAPI, Uvicorn, SQLAlchemy, and Google Generative AI SDK.

```
python -m venv venv  
source venv/bin/activate # On Windows: venv\Scripts\activate  
pip install -r requirements.txt
```

Configure environment variables

- Use environment variables to store sensitive data such as the Gemini API Key.
- These variables can be set directly in the terminal or added to a [.env](#) file which is loaded into the application using [python-dotenv](#).
`GOOGLE_API_KEY=your_gemini_api_key_here`

Frontend-Backend Rendering via Jinja2

All user interactions are facilitated via **Jinja2**-rendered templates:

- `index.html`: Collects user inputs
- `result.html`: Displays the generated AI plan
- `all_users.html`: Lists all user records for admin view

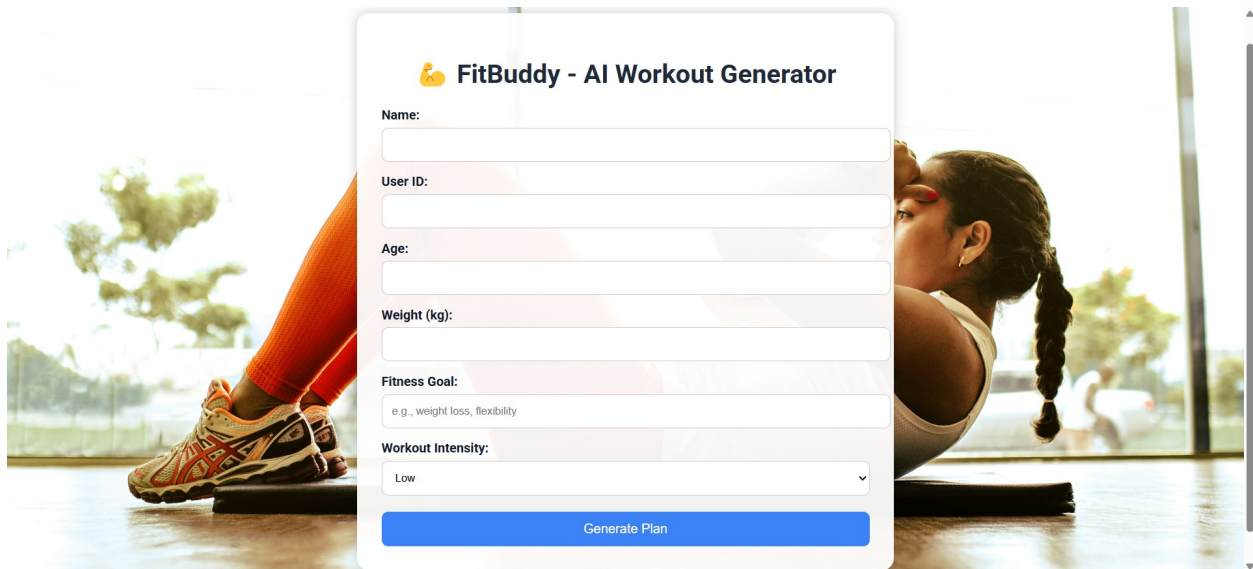
Activity 5.2: Testing and Verifying Local Deployment

Access the Application Locally

The FastAPI development server must be started using the command (**uvicorn main:app —reload**). Once the server is running, the application can be accessed through any web browser by navigating to `http://127.0.0.1:8000`. This URL loads the main user interface where individuals can input their details and generate personalized fitness plans. Additionally, the interactive API documentation provided by FastAPI can be viewed at `http://127.0.0.1:8000/docs`, allowing developers to explore and test available endpoints.

Exploring Website's web pages:

Home page:



The image shows a web form titled "FitBuddy - AI Workout Generator" overlaid on a background photograph of a person performing a plank exercise. The form is white with a light gray border and contains the following fields:

- Name:** A text input field.
- User ID:** A text input field.
- Age:** A text input field.
- Weight (kg):** A text input field.
- Fitness Goal:** A text input field with a placeholder "e.g., weight loss, flexibility".
- Workout Intensity:** A dropdown menu with "Low" selected.
- Generate Plan:** A blue button at the bottom of the form.

Description: The homepage of **FitBuddy – AI Workout Generator** features a clean and engaging user interface designed to provide a seamless experience for users seeking personalized fitness plans. The form prominently displayed in the center allows users to enter essential information including:

- **Name**
- **User ID**
- **Age**
- **Weight (kg)**
- **Fitness Goal** (e.g., weight loss, flexibility)
- **Workout Intensity** (Low, Medium, High)

Once users submit this data, they can click the **"Generate Plan"** button, which triggers the AI-powered backend to generate a customized workout and nutrition plan tailored to their goals and fitness profile. The visual appeal of the background image featuring an active individual reinforces the fitness-centric theme of the application, motivating users to take charge of their health journey.

This intuitive design ensures users of all experience levels can quickly get started with minimal effort.

Personalized workout generation page:

User Information: